3D-Belief: Embodied Belief Inference via Generative 3D World Modeling

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3d-belief.github.io

Abstract. Recent advances in visual generative models have highlighted the promise of learning generative world models. However, most existing approaches frame world modeling as novel-view synthesis or future-frame prediction, emphasizing visual realism rather than the structured uncertainty required by embodied agents acting under partial observability. In this work, we propose a different perspective: world modeling as embodied belief inference in 3D space. From this view, a world model should not merely render what may be seen, but maintain and update an agent's belief about the unobserved 3D world as new observations are acquired. We identify several key capabilities for such models, including spatially consistent scene memory, multi-hypothesis belief sampling, sequential belief updating, and semantically informed prediction of unseen regions. We instantiate these ideas in 3D-Belief, a generative 3D world model that infers explicit, actionable 3D beliefs from partial observations and updates them online over time. Unlike prior visual prediction models, 3D-Belief represents uncertainty directly in 3D, enabling embodied agents to imagine plausible scene completions and reason over partially observed environments. We evaluate 3D-Belief on 2D visual quality for scene memory and unobserved-scene imagination, object- and scene-level 3D imagination using our proposed 3D-CORE benchmark, and challenging object navigation tasks in both simulation and the real world. Experiments show that 3D-Belief improves 2D and 3D imagination quality and downstream embodied task performance compared to state-of-the-art methods.[*]

1. Introduction

Recent advances in video generation models have shown promising results on learning a generative world model that can predict future frames based on a single, multi-view, or streaming image inputs Song et al. [2025], Bar et al. [2025], Ren et al. [2025], Yu et al. [2024, 2025], Gu et al. [2025], Ma et al. [2025], Wu et al. [2025b], Huang et al. [2025], Cao et al. [2025]. These models have demonstrated the remarkable ability to render novel views of unobserved parts of the environments or the change of the seen parts of the environments caused by agents' actions.

However, there remains a substantial gap between what these models are trained to do and what a world model must represent for embodied agents operating from partial observations of a 3D scene. Predicting pixels is not the same as inferring the underlying 3D world: an agent in an unfamiliar environment must form a belief over both observed and unobserved regions, where this belief should capture task-relevant structure rather than only visual appearance. As new observations arrive, the

[*]Videos and demos are available at https://3d-belief.github.io/.

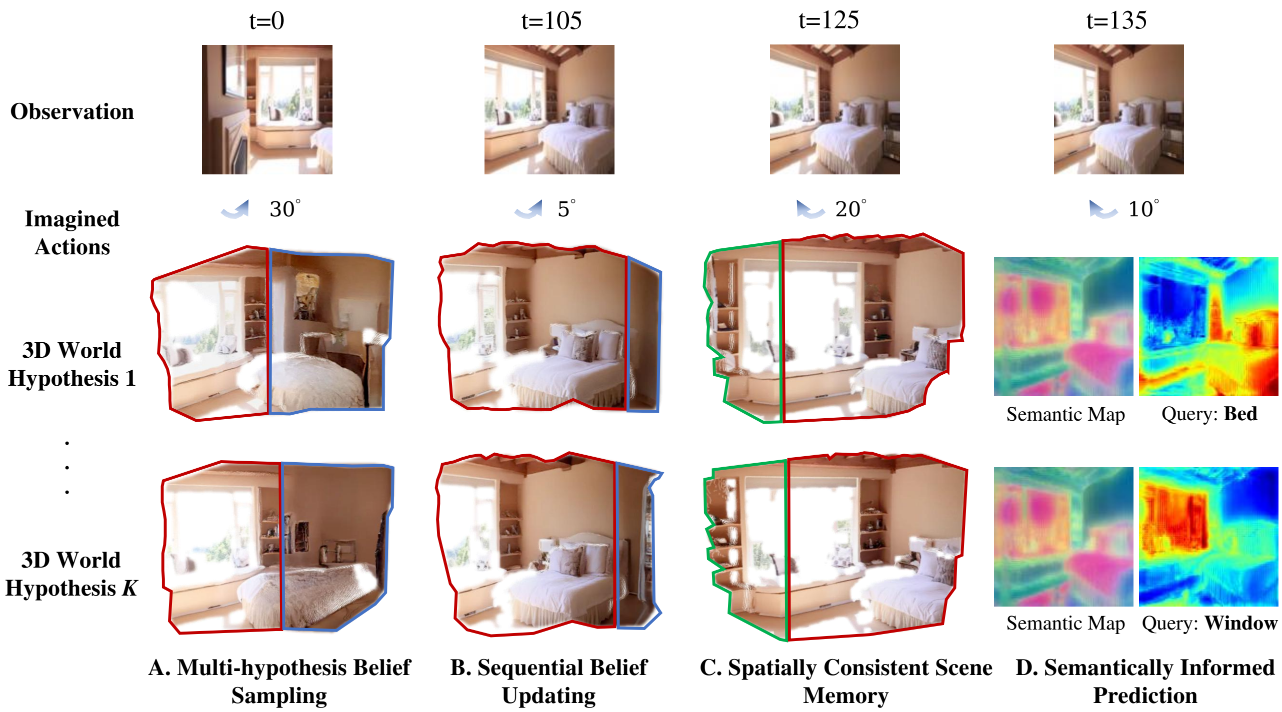

Figure 1: **Key capabilities of a generative 3D belief model.** An agent performs sequential mental simulation in an unknown environment. Top: selected RGB observations; second row: imagined actions; bottom: two sampled 3D world hypotheses. Regions supported by current observations are marked in red, imagined unseen regions in blue, and recalled content from scene memory in green. **A. Multi-hypothesis belief sampling:** at t=0, multiple plausible completions exist for unseen structure (e.g., unseen part of the bed and room). **B. Sequential belief updating:** incorporating observations from t=0 to 105 resolves ambiguity and updates the hypotheses (e.g., the appearance of the bed). **C. Spatially consistent scene memory:** revisiting viewpoints in mental simulation retrieves previously seen structure (e.g., bookshelf) with object permanence. **D. Semantically informed prediction:** semantic maps support task queries (e.g., *bed*, *window*).

agent should update this belief accordingly, forming an ever-evolving representation of the 3D world.

To achieve this, we theorize that robust embodied agents require a generative world model that has the following key capacities, as illustrated in Figure 1. First, the model needs to have a **spatially consistent scene memory**, which preserves the geometry and semantics of observed parts of the 3D world. Second, the model output should allow **multi-hypothesis belief sampling** to model uncertainty over the unseen parts of the 3D world. Third, the sequent nature of embodied tasks requires **sequential belief updating** conditioned on streaming partial observations. Lastly, it is crucial to produce **semantically informed belief prediction** which provides direct semantic prediction of the unseen parts of the 3D world to drive reasoning and planning.

As summarized in Table A1 in Appendix A1, existing methods cover only a subset of these capabilities. Pose- or action-conditioned video generation models Song et al. [2025], Wu et al. [2025a], Bar et al. [2025] provide strong 2D imagination from streaming observations, but remain in pixel space without explicit, actionable 3D beliefs. Methods that augment visual generation with 3D caches Ren et al. [2025], Yu et al. [2024, 2025], Zhou et al. [2025b] enable sequential scene memory, but mainly use 3D as a reconstruction cache rather than for diverse 3D imagination. Feedforward novel-view synthesis, reconstruction, and 3D foundation models Ye et al. [2024], Charatan et al. [2024], Chen et al. [2024], Jiang et al. [2025], Wang et al. [2024, 2025a], Zhuo et al. [2025], Wang et al. [2025b] maintain explicit 3D representations and may support online updates or limited inference of unseen structure, but lack uncertainty-aware multi-hypothesis 3D imagination. DFM Tewari et al. [2023] supports diverse, view-consistent imagination, but uses an implicit NeRF-style representation that is difficult to convert into explicit 3D structure for embodied tasks. Recent generative 3D world models Shen et al. [2026], Wang et al. [2026], Li et al. [2025] improve 3D-consistent generation, but are not designed for online belief inference from streaming egocentric observations. Finally, semantic grounding remains limited: while semantic 3D scene representations have been explored Ye et al. [2026], existing generative world models still lack open-vocabulary semantic predictions for unobserved 3D regions.

To bridge the gap, we propose ***3D-Belief***, a generative 3D world model that captures all the key abilities we hypothesized above. Built upon a diffusion model, 3D-Belief learns to predict explicit 3D representations of the full world state (including unseen regions) based on past egocentric visual observations. Critically, 3D-Belief can sequentially update its 3D prediction of the full world state based on new observations while maintaining consistency with all historical observations. The resulting 3D representations contain geometric, spatial, and semantic information about the 3D world, forming a coherent and queryable belief state that can support downstream embodied tasks.

To evaluate 3D-Belief, we assess its ability to predict and update 3D scene beliefs under partial observability. We first evaluate the 2D visual quality of the belief prediction, measuring both scene memorization in observed regions and imagination of unobserved regions from egocentric observations. We then introduce a novel benchmark, 3D-CORE (3D COntextual REasoning), to probe object- and scene-level 3D imagination. The proposed reasoning tasks provide a diagnostic evaluation of the four key capabilities of embodied agents' beliefs in 3D space. Finally, we evaluate whether the learned 3D scene beliefs can support downstream embodied tasks. In particular, we focus on open-vocabulary object navigation, where a robot is instructed to search for a target object in unseen environments. We test the model in both simulation and the real world with a mobile manipulator. Across these evaluations, 3D-Belief consistently improves 2D visual quality, explicit 3D imagination, and downstream embodied task performance compared to strong baselines.

In sum, our main contributions are: (1) a conceptual framework that models embodied belief inference via generative 3D world modeling; (2) 3D-Belief, a generative 3D world model that predicts actionable, semantic, and uncertainty-aware 3D scene beliefs from partial observations; (3) 3D-CORE, a benchmark for evaluating object- and scene-level 3D imagination in embodied settings; and (4) comprehensive evaluations demonstrating the effectiveness of 3D-Belief on 2D visual quality for scene memorization and imagination, 3D scene imagination, and downstream object navigation tasks in both simulation and the real world.

2. Related Work

Visual Generative Models. Recent video diffusion and view-synthesis models can generate realistic, high-resolution videos with controllable camera motion Song et al. [2025], Huang et al. [2025], Yu et al. [2024, 2025]. There has also been extensions to 3D/4D generation Zhen et al. [2025]. However, these models primarily generate frames in pixel space and typically require separate reconstruction modules to recover an explicit 3D scene representation. In contrast, we learn a generative model that directly predicts an explicit, actionable 3D belief.

World Model-Based Planning. Video world models have been used for action-conditioned rollouts to enable world model-based planning Du et al. [2023, 2024], Zhou et al. [2024], Zhang et al. [2024], Bar et al. [2025], Hafner et al. [2025]. Unlike these approaches, which plan over 2D rollouts, we show that a *3D* generative world model, such as 3D-Belief, can better support planning for embodied tasks in unseen 3D environments such as open-vocabulary object navigation.

Semantically-Embedded 3D Scene Representations. There has been work that augments 3D representations with semantics for open-vocabulary querying and task-directed reasoning. LERF associates text-aligned features with radiance fields for free-form language grounding Kerr et al. [2023], while ConceptGraph and DynaMem build persistent, online-updated spatial-semantic memories Gu et al. [2024], Liu et al. [2025]. These methods largely focus on representing what has been observed. In contrast, we introduce a semantically informed 3D belief that *predicts* unobserved regions and supports sequential updates.

3. 3D-Belief

In this work, we propose a generative 3D world model, 3D-Belief. As shown in Figure 2, it learns to predict and sequentially update explicit 3D representations of a scene online. We first formalize the 3D-Belief model in Section 3.1 and then introduce a model architecture to implement 3D-Belief in Section 3.2 - 3.4.

3.1. Formulation

We follow the definition of the belief in partially observable Markov decision making (POMDP) Kaelbling et al. [1998]. Specifically, belief $b(s^t)$ is a distribution of the state s^t at any given step t conditioned on past observations o^t, i.e., i.e., $b(s^t) = P(s^t|o^{1:t})$. Given new observations o^{t+1} at the recent step $t+1$, we can update the belief as

$$b(s^{t+1}) = \sum_{s^t} P(s^{t+1}|o^{t+1}, s^t)b(s^t). \tag{1}$$

Directly predicting the full 3D scene for the first (3D memory) and fourth (semantics) key capabilities can be extremely challenging. Therefore, we can instead predict a 3D representation of the scene, $z^t = \phi(s^t)$. Such representations (1) should preserve the important 3D structures and semantic information of the scene necessary for downstream embodied tasks, and (2) can be parametrized so that they can be conveniently constructed via model training. In this work, we consider 3D Gaussian Splatting (3DGS) Kerbl et al. [2023], as it not only provides an explicit 3D representation of a scene and can carry semantic feature embeddings in each primitive Qiu et al. [2024], but also allows fast rendering to support downstream embodied tasks.

Formally, $z^t = \{g_k\}_{k=1}^K$ with Gaussian primitives $g_k = (\mu_k, \Sigma_k, \alpha_k, S_k, e_k)$ (mean, covariance, opacity, SH appearance, semantic embedding). We split $z^t = z_o^t \cup z_i^t$ into observed and imagined Gaussians. For sequential belief updates, we rewrite Eq. equation 1 as an autoregressive update:

$$z^{t+1} \sim p(z^{t+1} \mid o^{t+1}, z_o^t). \tag{2}$$

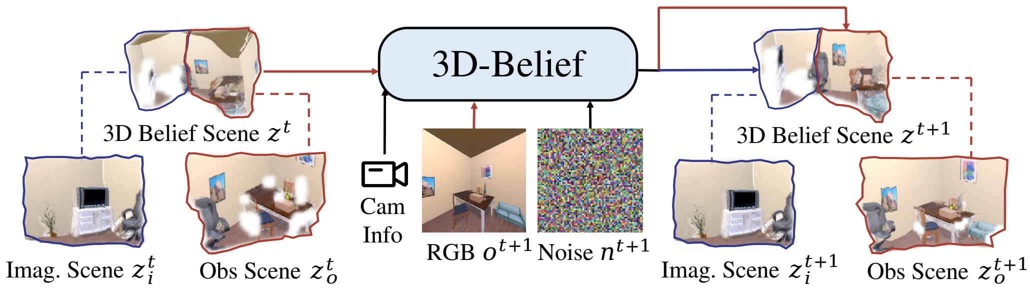

Figure 2: Overview of 3D-Belief. We represent the 3D representation of a sampled scene at step t as z^t, which includes the observed and imagined Gaussians z_o^t and z_i^t. Given z^t, a new observations o^{t+1}, and a sampled noise image n^{t+1}, 3D-Belief samples a new 3D scene representation at step $t+1$, i.e., z^{t+1}, to update the belief of the 3D world.

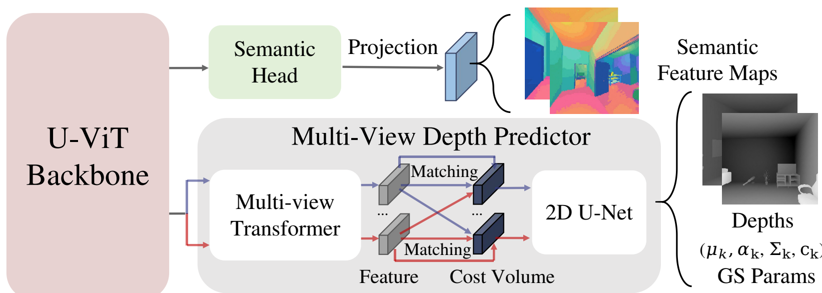

Figure 3: 3D-Belief model architecture: 3D Scene Diffusion.

In particular, z_o^t is expanded to form z_o^{t+1} by incorporating the new observation, whereas z_i^t is replaced with a new imagination, z_i^{t+1}, since the previously imagined content may conflict with new evidence. This formulation (i) enables a simple training procedure that requires supervision only over short horizons (as discussed in Sec. 3.2, we train our model using image pairs), (ii) supports test-time scaling for long-horizon planning via continual belief updates, and (iii) maintains constant per-step computational cost independent of the planning horizon.

Given z^t, we render an imagined view from any pose θ as $\hat{o} = \mathcal{G}(z^t, \theta)$ Kerbl et al. [2023], producing RGB, depth, and a semantic feature map from per-primitive embeddings. These renderings enable model-based mental simulation, and language queries over the semantic map let a planner score and select plans.

3.2. Model Architecture

Our model (Figure 3) uses a shared U-ViT backbone Song et al. [2025], Hoogeboom et al. [2024] with two heads that jointly predict a 3DGS scene with semantics. For geometric consistency, we employ an MVS-style 3DGS predictor with a multi-view Transformer and cost-volume module, in which the cost volume stores cross-view feature matching scores over discretized depth candidates to guide depth prediction, producing depths and Gaussian parameters that are lifted into primitives for fast differentiable rendering Chen et al. [2024]. A lightweight semantic head linearly projects backbone features into per-pixel semantic maps, trained by distillation from CLIP-style embeddings Kerr et al. [2023], enabling text-based querying at test time.

3.3. Diffusion Training

Unlike generative models that perform *frame-by-frame* diffusion in pixel or latent space, we adopt *scene-level 3D diffusion* as introduced in Tewari et al. [2023], applying diffusion to the entire 3D scene, i.e., the Gaussian primitives $z = g_{k_k}^{K}$ in this case. This design encourages global geometry modeling and multi-view consistency, yielding more effective 3D scene predictions for embodied tasks.

z is estimated by adding supervision with paired context images o^{ctxt} and target images o^{trgt}. The forward process at diffusion time step τ is defined as:

$$q(o_\tau^{\text{trgt}} \mid o_{\tau-1}^{\text{trgt}}) = \mathcal{N}(o_\tau^{\text{trgt}}; \sqrt{1-\beta_\tau} o_{\tau-1}^{\text{trgt}}, \beta_\tau I). \tag{3}$$

In the reverse process, we reconstruct o^{trgt} conditioned on o^{ctxt} and target camera parameters ϕ^{trgt} by predicting a denoised scene z_τ and rendering it into observation space via the GS renderer $\mathcal{G}(\cdot)$:

$$z_{\tau-1} = \Phi_\theta(o^{\text{ctxt}}, o_\tau^{\text{trgt}}; \tau, \phi^{\text{ctxt}}, \phi^{\text{trgt}}), \tag{4}$$

$$\hat{o}_{\tau-1}^{\text{trgt}} = \mathcal{G}(z_{\tau-1}, \phi^{\text{trgt}}). \tag{5}$$

Here, $\hat{o}_{t-1}^{\text{trgt}}$ serves as an estimate of the clean observation. Φ_θ is the neural network predicting 3D Gaussians, as defined in Sec 3.2. See Appendix A2 for the complete formulation.

3.4. Training Objective

We train Φ_θ with paired context and target observations. Let $\hat{o}^{\text{trgt}} = \mathcal{G}(z_0, \phi^{\text{trgt}})$ and $\hat{o}^{\text{ctxt}} = \mathcal{G}(z_0, \phi^{\text{ctxt}})$ be renderings of the predicted scene in the target and context cameras. For brevity, we use a view index $v \in \{\text{trgt}, \text{ctxt}\}$ and let $o^v \in \{o^{\text{trgt}}, o^{\text{ctxt}}\}$ and $\hat{o}^v \in \{\hat{o}^{\text{trgt}}, \hat{o}^{\text{ctxt}}\}$ denote the corresponding observation and rendering in view v. We use $I(\cdot)$ to denote RGB channels.

RGB loss. The RGB loss is $\mathcal{L}_{\text{rgb}} = \sum_{v\in\{\text{trgt},\text{ctxt}\}} \|I(\hat{o}^v) - I(o^v)\|_2^2$.

Semantic loss. We align the rendered semantic feature map with features extracted from image patches. Let $S(\hat{o}^v) \in \mathbb{R}^{H\times W\times d}$ be a per-pixel semantic feature map rendered from the predicted scene in view v. For each v, we sample patch centers $\mathcal{P}^v = \{\mathbf{u}_j\}_{j=1}^M$ on $I(o^v)$, crop patches $\pi(o^v, \mathbf{u}_j)$, and compute features using a frozen CLIP Ilharco et al. [2021] image encoder $f_{\text{clip}}(\cdot)$. We supervise the semantic feature at the corresponding pixel using $\mathcal{L}_{\text{sem}} = \sum_{v\in\{\text{trgt},\text{ctxt}\}} \frac{1}{M} \sum_{j=1}^M \left\| S(\hat{o}^v)(\mathbf{u}_j) - f_{\text{clip}}\big(\pi(o^v, \mathbf{u}_j)\big) \right\|_2^2$.

Depth loss (optional). When ground-truth depth is available, we additionally supervise the rendered depth. Let $D(\cdot) \in \mathbb{R}^{H\times W}$ denote the depth channel of an observation. For each view v, define $\hat{d^v} = D(\hat{o}^v)$ and $d^v = D(o^v)$. To handle missing or invalid depth values, let $m^v \in \{0,1\}^{H\times W}$ be a validity mask, where 1 indicates valid depth. We use a masked ℓ_1 loss: $\mathcal{L}_{\text{depth}} = \sum_{v\in\{\text{trgt},\text{ctxt}\}} \frac{1}{\sum_{\mathbf{u}} m^v(\mathbf{u})} \sum_{\mathbf{u}} m^v(\mathbf{u}) \left| \hat{d^v}(\mathbf{u}) - d^v(\mathbf{u}) \right|$.

4. Experiments

We conduct a comprehensive evaluation of 3D-Belief as a generative 3D world model for embodied belief inference under partial observability. Our experiments evaluate its capabilities across 2D

Table 1: **2D visual quality of belief prediction.** Scene memory is evaluated with paired metrics, while scene imagination is evaluated with distributional metrics. Best results are highlighted in **bold**.

Method	LPIPS↓	PSNR↑	SSIM↑
NWM	0.1876	18.75	0.702
DFoT	0.1206	23.35	0.841
Ours	**0.0502**	**28.81**	**0.928**

(a) Observed scene

Method	FVD↓	FID↓
NWM	487.4	89.28
DFoT	429.7	72.82
Ours	**271.8**	**47.24**

(b) Imagined scene

visual quality of belief prediction, 3D object and scene imagination, and belief-guided planning with inference-time scaling. Appendix A4 provides more implementation details.

4.1. Experiment 1: 2D Visual Quality of Belief Prediction

Embodied belief requires both faithful memorization of observed regions and plausible imagination of unobserved space. One way to evaluate these capacities is to examine the visual quality of the belief predictions, similar to the common evaluations for video generation models Ren et al. [2025]. A successful model needs to provide visual rendering of imagination that not only have rich visual details but also preserves the geometric and semantic integrity of the 3D scene.

Evaluation protocol. We use a two-part evaluation protocol. For observed regions, we condition the model on both ends of a trajectory and predict the intermediate novel views. Since these target views have paired ground truth, we evaluate them using LPIPS Zhang et al. [2018], PSNR, and SSIM. For unobserved regions, we condition the model only on the starting frame and generate views beyond the observed field of view. Because these predictions are inherently multimodal, we report distributional metrics, FVD Unterthiner et al. [2019] and FID Heusel et al. [2017], to measure the realism and consistency of the generated visual distribution. We evaluate primarily on navigation datasets generated in AI2-THOR Kolve et al. [2017] following Ehsani et al. [2024], as this setting directly tests embodied belief prediction under egocentric exploration and partial observability. On AI2-THOR, we compare against DFoT Song et al. [2025] and NWM Bar et al. [2025] fine-tuned on the AI2-THOR training set. We also evaluate on RealEstate10K Zhou et al. [2018] in Appendix A3.1.2 as a complementary real-world visual prediction benchmark, comparing against pretrained DFoT Song et al. [2025] and GEN3C Ren et al. [2025]. Qualitative results from both datasets are shown in Figure 4.

Results. Table 1 shows that 3D-Belief consistently outperforms the baselines across both evaluation settings. On observed-view reconstruction, our method achieves the lowest LPIPS and the highest PSNR and SSIM, indicating that it better preserves the geometry and appearance of seen scene content when synthesizing intermediate views. On scene imagination, 3D-Belief also achieves the lowest FVD and FID, suggesting that explicit 3D belief modeling improves the realism and temporal consistency of predictions beyond the observed field of view. Complementary results on RealEstate10K further show that 3D-Belief achieves substantially lower FID and FVD than pretrained video prediction baselines, indicating strong distributional visual imagination on real-world trajectories (see Appendix A3.1.2).

Figure 4 highlights two common baseline failure modes. First, baselines often exhibit *semantic drift* beyond observed views. DFoT turns grass into a solid green ground in RealEstate10K, while AI2-THOR baselines predict opened doors without preserving plausible opening structure. In contrast, 3D-Belief better maintains semantic and scene consistency. Second, baselines suffer from *pose and*

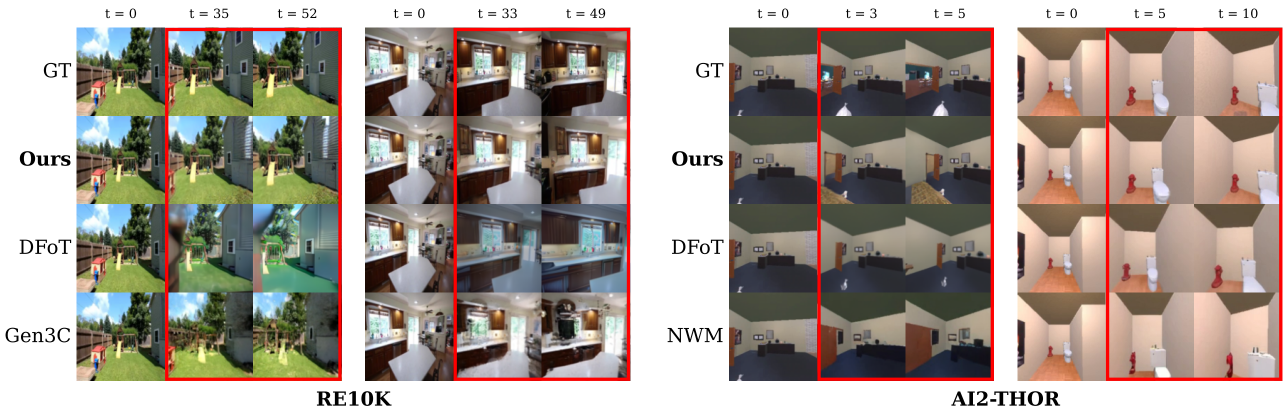

Figure 4: **Qualitative results of 2D visual quality of belief prediction.** Model predictions are highlighted with red boxes. We compare 3D-Belief against baselines on the 2D visual quality of imagination. 3D-Belief better preserves pose alignment, geometry, and texture details. See Appendix A3.1.1 for more qualitative results.

geometric drift under viewpoint changes, causing generated content to shift or distort relative to the target view. This is especially visible in the right-hand AI2-THOR example, where object shapes and room layout degrade over time. 3D-Belief better preserves pose alignment and rigid scene structure through its explicit 3D belief representation.

4.2. Experiment 2: 3D Object and Scene Imagination

4.2.1. 3D-CORE Benchmark

In embodied tasks, agents must integrate partial observations over time, infer unobserved regions, and maintain stable beliefs while moving. For instance, object search requires reasoning about targets hidden behind occluders, while navigation benefits from inferring plausible free space and room layout from limited views. However, common evaluations of generative world models often focus on visual fidelity or short-horizon prediction Song et al. [2025], Bar et al. [2025], as in Sec. 4.1. These metrics do not directly test whether a model forms consistent 3D beliefs under partial observability. We therefore introduce 3D-CORE (3D COntextual REasoning), a benchmark for embodied belief reasoning in 3D space. 3D-CORE tests: (1) **spatial expansion** beyond observed regions, (2) **semantic reasoning** grounded in 3D structure, and (3) **long-horizon consistency** under large viewpoint changes.

Benchmark Construction. All Scenarios are built in AI2-THOR Kolve et al. [2017] with Proc-THOR Deitke et al. [2022] houses with diverse layouts and rich object assets from Objaverse Deitke et al. [2023]. We render egocentric RGB streams with known camera poses and provide ground-truth 3D geometry and semantics for evaluation.

Tasks and Metrics. 3D-CORE contains three complementary tasks that isolate distinct but essential aspects of contextual 3D reasoning. (See Appendix A4.3.1 for definitions of all metrics.)

Task 1: Object Completion (233 tasks): The model observes a partially visible target object and completes its 3D geometry and appearance, producing plausible shape/texture consistent with the object category. Metrics: BEV IoU, 3D IoU, Chamfer distance, SigLIP similarity, and Recognition.

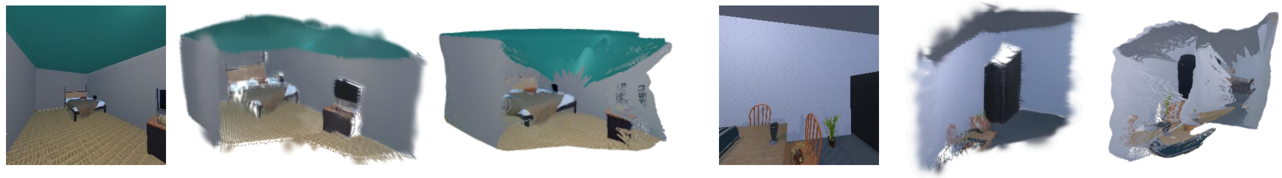

Figure 5: **Qualitative results of 3D object imagination.** Compared with the tested baseline, 3D-Belief better infers object-level semantic and geometric structure from the observed image, enabling more complete and consistent 3D imagination.

Table 2: Results on 3D-CORE: Object Completion, Room Completion, and Object Permanence.

	Object Completion (55% Visibility)					Room Completion			Object Permanence	
Method	**BEV IoU ↑**	**3D IoU↑**	**Chamfer↓**	**SigLIP↑**	**Recognition↑**	**Obj. F1 ↑**	**Occ. Acc. ↑**	**Occ. IoU ↑**	**LPIPS ↓**	**SigLIP ↑**
DFoT-VGGT	0.362	0.243	0.830	0.798	0.767	0.531	0.252	0.110	0.555	0.907
3D Belief	**0.484**	**0.318**	**0.216**	**0.855**	**0.930**	**0.536**	**0.900**	**0.442**	**0.123**	**0.978**

Task 2: Room Completion (263 tasks): Given a single egocentric view of a room (with pose), the model predicts the unseen layout and semantics to support planning (e.g., plausible navigable regions). Metrics: Object Prediction F1, occupancy accuracy (known cells), and occupancy IoU.

Task 3: Object Permanence (474 tasks): The model is rolled out along a trajectory with large viewpoint changes that returns to the start. We test whether the 3D belief stays stable, meaning that objects should not drift, or change identity upon revisiting. Metrics: LPIPS and SigLIP similarity.

4.2.2. Evaluation on 3D-CORE

We evaluate 3D-Belief's ability to imagine object- and scene-level 3D structure on the 3D-CORE benchmark. Given an initial RGB observation and an imagination camera trajectory, the agent predicts a completed 3D belief conditioned on the observation. Metrics are computed using semantics, geometry, or rendered views extracted from the predicted 3D representation.

Baselines. To the best of our knowledge, no existing method can directly imagine an explicit 3D scene. Thus, we construct a strong baseline, DFoT-VGGT, by combining two SOTA models. DFoT-VGGT is a "imagination-then-lift" baseline that first uses DFoT Song et al. [2025] to generate imagined observations and then applies VGGT Wang et al. [2025a] to lift these predictions into a 3D representation.

Results. Table 2 summarizes the main results. Overall, 3D-Belief consistently outperforms DFoT-VGGT across all three 3D-CORE tasks, demonstrating stronger 3D belief reasoning for object completion, room-level spatial inference, and long-horizon consistency.

On **Object Completion** (full results in Appendix A3.3.1), 3D-Belief improves both geometry and semantics across visibility levels, with higher BEV/3D IoU, lower Chamfer distance, and better SigLIP similarity and VLM recognition. These gains indicate more faithful 3D completion and stronger preservation of object appearance and category semantics.

On **Room Completion**, the two models achieve similar object prediction F1, but 3D-Belief produces substantially more accurate occupancy beliefs, improving known-cell accuracy, free/occupied IoU, and overall occupancy IoU. This suggests stronger spatial belief estimation for downstream tasks.

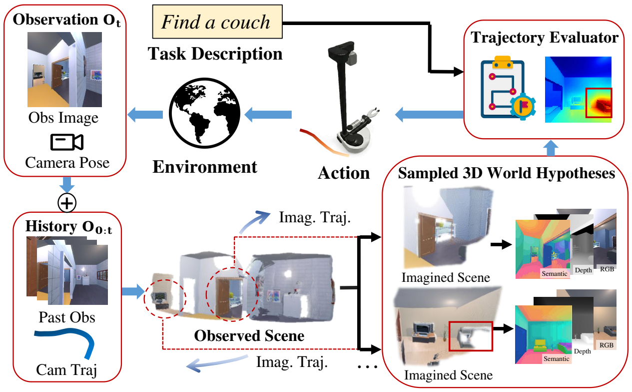

Figure 6: **Belief-guided planning with 3D-Belief**. Dotted circles are imagination regions for visualization, and the red boxes are the potential target objects in imagined renders.

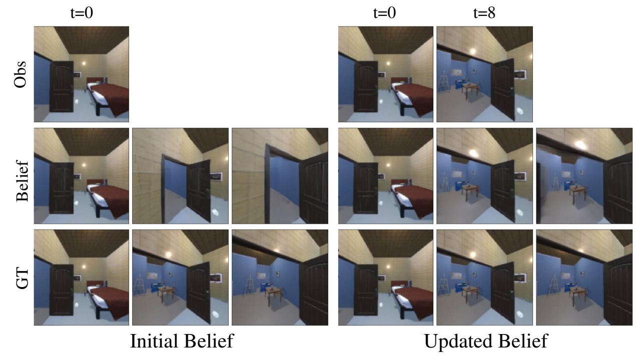

Figure 7: **Belief updating during navigation**. The initial belief is formed from the first observation, while the updated belief incorporates later views, revising imagined regions and supporting navigation decisions.

On **Object Permanence**, 3D-Belief shows better long-horizon consistency under large camera motions, achieving higher perceptual and semantic consistency when returning to the initial viewpoint. This indicates reduced drift and more stable geometry and object identity over extended rollouts.

4.3. Experiment 3: Belief-Guided Planning with Inference-Time Scaling

Embodied planning under partial observability requires an agent to act on both observed evidence and plausible unseen scene structure. We evaluate whether 3D-Belief can serve as a mental 3D world model for navigation. As shown in Figure 6, at each step, the agent updates its belief from egocentric RGB history and camera poses, samples multiple plausible 3D beliefs, and evaluates candidate paths by rendering imagined observations along them. Rollouts are scored with open-vocabulary semantic queries on the imagined belief, guiding the planner toward paths likely to reveal or approach the target. Figure 7 illustrates how 3D-Belief updates the agent's belief during navigation. Additional implementation details are provided in Appendix A4.2.

For simulation, we train 3D-Belief on AI2-THOR Kolve et al. [2017] navigation episodes; for real-world experiments, we train a separate model on RealEstate10K Zhou et al. [2018]. To support open-vocabulary goals, semantic heads are trained with OpenCLIP Ilharco et al. [2021] features and regularized with DINOv3 Siméoni et al. [2025], following LERF Kerr et al. [2023]. This enables *inference-time scaling*: the agent samples multiple beliefs, evaluates multiple imagined paths, and selects the rollout with the strongest goal progress under uncertainty.

4.3.1. Object Navigation in Simulations

Simulation Environment. We conduct all simulated tasks following Ehsani et al. [2024] in the AI2-THOR simulator Kolve et al. [2017], using ProcTHOR house assets Deitke et al. [2022]. Unless otherwise noted, we evaluate on 135 unseen houses for test, ensuring generalization to novel layouts and object configurations.

Task Definition. We define the task following Ehsani et al. [2024]. Each episode starts the agent at a

Table 3: **Object navigation results in simulations and the real world.** More results are provided in Appendix A3.2.

Metric	Simulations								Real World	
	VGGT (w/ frontier)	VGGT (w/ Gemini 3.0)	DFoT-VGGT (w/ Gemini 3.0)	NWM-VGGT (w/ Gemini 3.0)	Gemini 3.0	Qwen3-VL-8B Instruct	SPOC	**Ours**	Gemini 3.0	**Ours**
SR% ↑	27.50	25.00	26.05	25.00	45.00	18.33	31.67	**59.17**	23.08	**55.56**
SPL% ↑	25.82	24.10	24.59	23.46	37.81	14.08	30.97	**39.07**	–	–
SEL% ↑	22.66	22.66	23.43	20.76	**41.47**	16.94	30.56	40.24	13.55	**35.91**
Token (/step) ↓	**0**	1448.02	1210.08	552.12	7512.70	220.29	**0**	**0**	2317.09	**0**

random location with a target object category, which may require multi-room exploration. The agent receives egocentric RGB observations and poses sequentially and must plan actions online. Success requires the agent to be within a proximity threshold of the target and to see it in the central region of view.

Metrics. We evaluate task performance using Success Rate (SR) for overall success, Success-weighted by Path Length (SPL) for path efficiency, Success-weighted by Episode Length (SEL) Eftekhar et al. [2023], Yokoyama et al. [2021] for time efficiency. We also report token costs for VLM usage.

Baselines. We compare 3D-Belief against four categories of methods, as shown below:

- 3D Reconstruction. This group tests whether high-quality **reconstruction of only the observed geometry** is sufficient for efficient search. We use VGGT Wang et al. [2025a] to build a 3D cache from the agent's RGB history. Variants differ only in the goal-selection module: a classical frontier-based explorer (VGGT w/ frontier) and VLM-driven waypoint selection (VGGT w/ GPT-5 mini or Gemini 3.0).
- Lifted 2D Imagination. This group tests whether imagined future observations can be lifted into 3D for downstream navigation. We first use video prediction models, DFoT and NWM, to imagine future RGB observations, and then use VGGT to reconstruct a 3D representation from the generated views. Goal selection is performed by a VLM, either GPT-5 mini or Gemini 3.0.
- VLM Agents. This group evaluates whether end-to-end VLM-based agents can directly solve object navigation from visual observations. We compare against GPT-5 mini, Gemini 3.0, and Qwen3-VL-8B-Instruct, which select actions or subgoals based on the agent's observation history.
- Policy Model. This group evaluates a learned navigation policy trained for embodied object search. We include SPOC Ehsani et al. [2024], which provides a strong policy-based baseline without explicit VLM reasoning during online inference.

Results. Table 2 summarizes the quantitative results, and Figure 5 shows qualitative examples of object-level 3D imagination. Overall, 3D-Belief consistently outperforms DFoT-VGGT across all three 3D-CORE tasks, demonstrating stronger 3D belief reasoning for object completion, room-level spatial inference, and long-horizon consistency.

4.3.2. Object Navigation in the Real World

We evaluate 3D-Belief on a real mobile manipulation platform (Hello Robot Stretch) in a mock apartment environment. The environment contains typical household furniture and objects, forming realistic object navigation scenarios. See Appendix A4.2.2 for an example environment setup. Note that the environments, objects, and target descriptions are all unseen during training, making this a real-world open-vocabulary object navigation setting.

Success Criteria. At the beginning of each episode, the robot is given the name of the target object (e.g., red mug), and must explore the environment and stop when it is found. An episode is marked as successful if (1) the robot reaches within a distance threshold of the target and (2) the target is within the central region of the egocentric view with a facing angle within a tolerance.

Baseline. We test the best performing baseline in simulations, VLM agent based on Gemini 3.0.

Result. Results are shown in Table 3. Across real-robot episodes, 3D-Belief enables more reliable and efficient performance than the Gemini 3.0 agent. It achieves a higher success rate and improved efficiency, demonstrating that explicit, online-updatable 3D beliefs transfer to real-world settings and can robustly support embodied decision making under sensor noise.

5. Conclusion

In this work, we proposed a new framework for embodied belief inference via generative 3D world modeling and identified key capabilities for practical 3D belief modeling. We then proposed 3D-Belief, which predicts unseen regions in an explicit 3D representation from partial observations and updates this belief online. Experimental results on 2D visual quality, contextual reasoning, and object navigation have demonstrated that 3D-Belief improves upon existing generative world models in 2D and 3D scene memory and imagination, as well as downstream embodied task performance.

Limitations and Future Work. Our 3D-Belief model assumes a static world. Future work will incorporate dynamic world modeling to support broader embodied tasks. Another direction is to improve the controllability of 3D imagination by conditioning hypothesis sampling on high-level guidance, such as language descriptions or scene graphs of the imagined 3D world.

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Appendix

A1. Extended Related Work

Table A1: Comparison of generative world model capabilities. We use ✓ to indicate the capability is supported and ✗ otherwise. "Scene Mem." denotes memory of observed scene regions; "2D Imag." denotes pixel-space multi-hypothesis imagination; "3D Imag." denotes multi-hypothesis imagination of explicit 3D representations beyond observed geometry; "Sequential" denotes online updates from streaming observations; and "Semantic" denotes semantic features or labels.

Models	Scene Mem.	2D Imag.	3D Imag.	Sequential	Semantic
DFoT Song et al. [2025]	✗	✓	✗	✓	✗
NWM Bar et al. [2025]	✗	✓	✗	✓	✗
Persistent EWM Zhou et al. [2025b]	✓	✓	✗	✓	✗
GEN3C Ren et al. [2025]	✓	✓	✗	✓	✗
ViewCrafter Yu et al. [2024]	✓	✓	✗	✗	✗
TrajectoryCrafter Yu et al. [2025]	✓	✓	✗	✗	✗
MVSplat Chen et al. [2024]	✓	✗	✗	✗	✗
VGGT Wang et al. [2025a]	✓	✗	✗	✗	✗
CUT3R Wang et al. [2025b]	✓	✗	✗	✓	✗
DFM Tewari et al. [2023]	✓	✓	✗	✓	✗
Lyra 2.0 Shen et al. [2026]	✓	✓	✓	✗	✗
Rein3D Wang et al. [2026]	✓	✓	✓	✗	✗
FlashWorld Li et al. [2025]	✗	✓	✓	✗	✗
SemGS Ye et al. [2026]	✓	✗	✗	✗	✓
3D-Belief	✓	✓	✓	✓	✓

A2. Diffusion Training Formulation

The forward process is defined as:

$$q(o_\tau^{\text{trgt}} \mid o_{\tau-1}^{\text{trgt}}) = \mathcal{N}(o_\tau^{\text{trgt}}; \sqrt{1-\beta_\tau} o_{\tau-1}^{\text{trgt}}, \beta_\tau I). \tag{A6}$$

In the reverse process, we reconstruct o^{trgt} conditioned on o^{ctxt} and camera parameters ϕ^{trgt}:

$$\begin{aligned} & p_\theta(o_{0:T}^{\text{trgt}} \mid o^{\text{ctxt}}; \phi^{\text{ctxt}}, \phi^{\text{trgt}}) && \text{(A7)} \\ & = p(o_T^{\text{trgt}}) \prod_{\tau=0}^{T} p_\theta(o_{\tau-1}^{\text{trgt}} \mid o_\tau^{\text{trgt}}, o^{\text{ctxt}}; \phi^{\text{ctxt}}, \phi^{\text{trgt}}), && \text{(A8)} \end{aligned}$$

where $p_\theta(o_{\tau-1}^{\text{trgt}} \mid o_\tau^{\text{trgt}}, o^{\text{ctxt}}; \phi^{\text{ctxt}}, \phi^{\text{trgt}})$ is implemented by first predicting an intermediate clean 3DGS scene z_τ and then mapping it to a denoised observation using the Gaussian Splatting rendering function $\mathcal{G}(\cdot)$:

$$\begin{aligned} z_{\tau-1} &= \Phi_\theta(o^{\text{ctxt}}, o_\tau^{\text{trgt}}; \tau, \phi^{\text{ctxt}}, \phi^{\text{trgt}}), && \text{(A9)} \\ \hat{o}_{\tau-1}^{\text{trgt}} &= \mathcal{G}(z_{\tau-1}, \phi^{\text{trgt}}), && \text{(A10)} \\ o_{\tau-1}^{\text{trgt}} &\sim \mathcal{N}(o_{\tau-1}^{\text{trgt}}; C_{\tau-1}\hat{o}_{\tau-1}^{\text{trgt}}, \hat{\beta}_\tau I). && \text{(A11)} \end{aligned}$$

Here, $\hat{o}^{\text{trgt}}_{\tau-1}$ serves as an estimate of the clean observation. The constants $C_{\tau-1}$ and $\hat{\beta}_{\tau-1}$ are specifically chosen to ensure the noise at time $\tau-1$ aligns with the total noise introduced during the forward process. Φ_θ is a neural network shown in Figure 3. During test time, the scene is generated by iteratively applying Eqs. equation A9-equation A11, starting from an initial distribution $p(o^{\text{trgt}}_{\tau=T}) \sim \mathcal{N}(0, I)$.

The model defines a generative process over the Gaussian primitives, formalized as:

$$p_{\theta,\phi^{\text{trgt}}}(z_{0:T} \mid o^{\text{ctxt}}; \phi^{\text{ctxt}}) = \prod_{\tau=1}^{T} p_\theta(z_{\tau-1} \mid o^{\text{trgt}}_\tau, o^{\text{ctxt}}; \phi^{\text{ctxt}}, \phi^{\text{trgt}}). \tag{A12}$$

A3. Extended Results

A3.1. Extended Vision Results

A3.1.1. More Qualitative Results

We include more qualitative results in this section.

A3.1.2. More Quantitative Results

We additionally evaluate 3D-Belief on RealEstate10K Zhou et al. [2018] as a complementary real-world visual prediction benchmark. Unlike AI2-THOR, which directly matches our embodied setting with egocentric navigation trajectories and partially observed indoor scenes, RealEstate10K consists of real-world camera trajectories collected from real-estate videos. This benchmark therefore emphasizes photorealistic novel-view synthesis and video-level visual fidelity rather than explicit 3D belief inference, online belief updating, or downstream embodied utility.

Following the evaluation protocol in Sec. 4.1, we evaluate scene memory using paired reconstruction metrics, including PSNR, SSIM, and LPIPS, and evaluate scene imagination using distributional metrics, including FID and FVD. We compare 3D-Belief against pretrained DFoT Song et al. [2025] and GEN3C Ren et al. [2025].

Table A2: **Complementary visual prediction results on RealEstate10K.** RealEstate10K emphasizes photorealistic novel-view synthesis from real-world video trajectories. Best results are highlighted in **bold**.

Method	Obs PSNR↑	Obs SSIM↑	Obs LPIPS↓	Img FID↓	Img FVD↓
Ours	20.010	0.654	0.1410	**24.817**	**55.910**
DFoT Song et al. [2025]	22.395	0.717	0.1213	37.447	127.024
GEN3C Ren et al. [2025]	**22.901**	**0.775**	**0.0975**	51.918	175.453

As shown in Table A2, 3D-Belief shows different behavior on scene memory and scene imagination. On paired scene memory metrics, GEN3C achieves the best PSNR, SSIM, and LPIPS, indicating stronger image-level fidelity for observed-view interpolation on RealEstate10K. In contrast, 3D-Belief achieves the lowest FID and FVD by a large margin, suggesting that its generated unobserved scenes

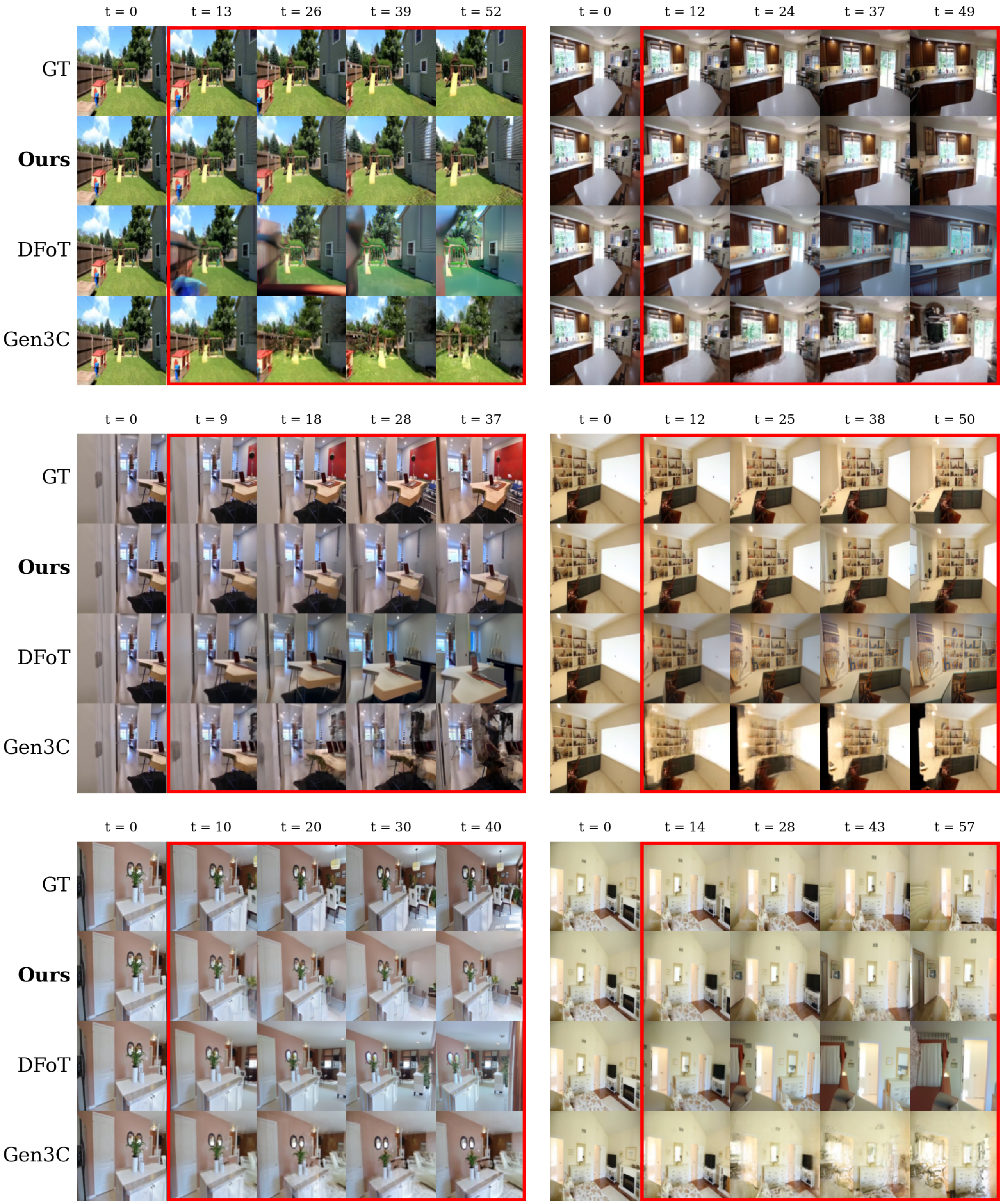

Figure A1: Qualitative results on RE10K.

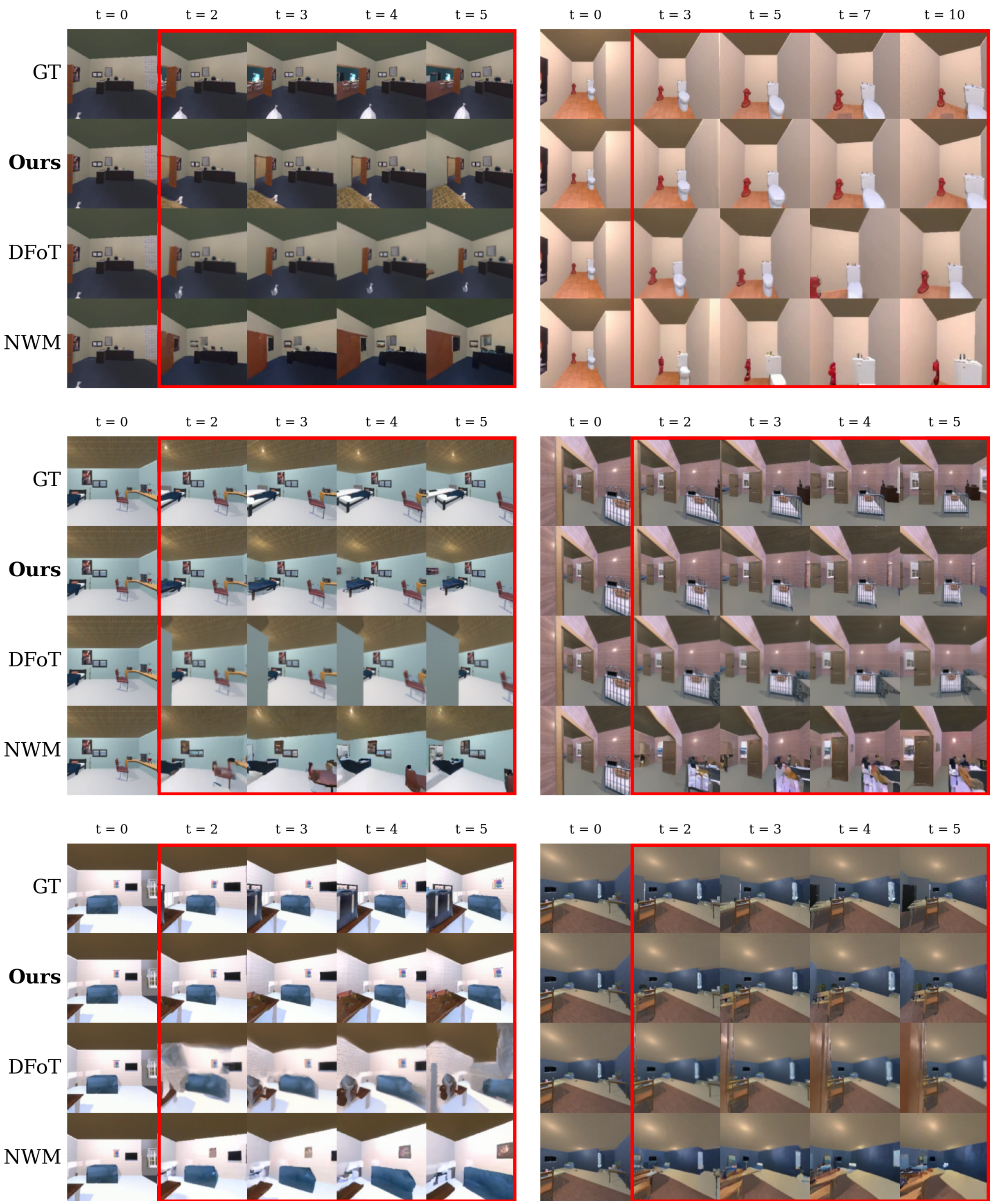

Figure A2: Qualitative results on AI2THOR.

better match the real visual distribution and maintain stronger video-level consistency. This result supports the role of explicit 3D belief modeling for visual imagination, especially when predictions are inherently multimodal and cannot be judged only by paired pixel-level metrics.

We therefore treat RealEstate10K as a complementary visual prediction benchmark rather than the primary embodied belief evaluation setting. The main claims of 3D-Belief are better reflected in AI2-THOR, 3D-CORE, and downstream embodied navigation, where partial observability, explicit 3D belief quality, semantic structure, and online belief updating can be evaluated more directly. Nevertheless, the low FID and FVD on RealEstate10K indicate that 3D-Belief also provides strong distributional visual imagination on real-world video trajectories.

A3.2. Extended Planning Results

A3.2.1. Full Navigation Results

We report the full object navigation results in Table A3, including additional baselines and inference speed comparisons.

A3.2.2. Ablation Studies

We report ablations for the 3D-Belief model in the simulated object navigation task in Table A4.

A3.2.3. Statistical Variation

In Table A5, we present per-episode variability for simulated object navigation, comparing our 3D-Belief model with the strongest baseline, Gemini 3.0. We report 95% confidence intervals computed over episodes, using Wilson binomial intervals for success rate and percentile bootstrap intervals for all other episode-level metrics.

A3.2.4. Use VLMs for Semantics

To study the impact of the built-in open-vocabulary semantics in 3D-Belief, we performed extra experiments with VLMs as the model semantic support. Results are reported in Table A6. We find that with VLMs, 3D-Belief model can perform better in navigation tasks, indicating that VLMs can help improve the semantic performance in terms of embodied decision making (e.g. selecting among different candidate paths). The performance of the built-in semantic module is largely affected by the CLIP-style vision-language alignment foundation models.

A3.3. Full Results on 3D-CORE

A3.3.1. Object Completion

We report the full object completion results with different visibilities in Table A7.

Table A3: **Object navigation results in simulations and the real world.** Higher is better for SR, SPL, and SEL; lower is better for inference time and VLM tokens. Best results are highlighted in **bold**, and second-best simulation results are underlined.

Model	SR% ↑	SPL% ↑	SEL% ↑	Inf. (s/step) ↓	Token (/step) ↓
Simulations					
VGGT (w/ frontier)	27.50	25.82	22.66	25.13	**0**
VGGT (w/ GPT-5m)	25.00	24.17	22.40	17.92	468.66
VGGT (w/ Gemini 3.0)	25.00	24.10	22.66	18.87	1448.02
DFoT-VGGT (w/ GPT-5m)	21.01	20.73	20.48	32.80	423.86
DFoT-VGGT (w/ Gemini 3.0)	26.05	24.59	23.43	45.57	1210.08
NWM-VGGT (w/ GPT-5m)	25.00	23.35	23.00	62.75	315.28
NWM-VGGT (w/ Gemini 3.0)	25.00	23.46	20.76	54.41	552.12
GPT-5m	23.33	17.66	20.19	11.10	6644.90
Gemini 3.0	<u>45.00</u>	<u>37.81</u>	**41.47**	7.58	7512.70
Qwen3-VL-8B-Instruct	18.33	14.08	16.94	2.30	<u>220.29</u>
SPOC	31.37	30.97	30.56	**0.22**	**0**
3D-Belief	**59.17**	**39.07**	<u>40.24</u>	<u>2.14</u>	**0**
Real World					
Gemini 3.0	23.08	–	13.55	4.52	2317.09
3D-Belief	**55.56**	–	**35.91**	**2.02**	**0**

A3.3.2. Room Completion

We report room completion results with more metrics in Table A8.

A3.4. Spatial Reasoning QA

A3.4.1. Evaluation on SAT-Real

To compare 3D-Belief with prior world models on the benefits they provide for reasoning about motion and space, we evaluate on the SAT-Real benchmark, which probes VLM reasoning by presenting two RGB images and asking binary questions over multiple subsets, namely Egocentric Movement (EM), Object Movement (OM), Egocentric-Allocentric Perspective Taking (Pers), Goal Aiming (GA), and Egocentric Action Consequence (EA).

Baselines. We follow MindJourney Yang et al. [2025b] to perform test-time scaling with a world

Table A4: Ablation studies for simulated object navigation tasks.

Models	SR% ↑	SPL% ↑	SEL% ↑	Token (/step) ↓
3D-Belief	45.83	36.43	32.99	0
w/o geometry	17.50	13.25	14.75	0
single hypothesis	35.14	28.85	26.81	0

Table A5: Episode-level performance for simulated object navigation. Values are means with 95% confidence intervals in brackets.

Models	SR% ↑	SPL% ↑	SEL% ↑
Strongest baseline	45.00 [36.39, 53.92]	37.81 [29.55, 46.18]	41.47 [33.12, 50.02]
3D-Belief	59.17 [50.22, 67.54]	39.07 [31.50, 46.87]	40.24 [32.70, 47.65]

model: given the current observation, we use beam search to simulate plausible camera motions and generate additional visual evidence using the world model, which is then provided to the VLM as context for answering questions. We compare 3D-Belief against two video world models, **SWM** Yang et al. [2025b] and **SVC** Zhou et al. [2025a], using the same inference protocol and budget (same beam width and rollout length). All methods are paired with the same VLM backbone (Gemini-3.0), and differ only in the auxiliary world model used to generate the additional context.

Results. Table A9 shows that augmenting Gemini-3.0 with 3D-Belief yields the best overall performance (88.7%), outperforming both the base VLM (85.3%) and the best video-world-model baseline (86.7%). The gains are concentrated on subsets that require reasoning about camera motion and 3D space: 3D-Belief improves EM to 100.0% (vs. 95.7%), and substantially boosts EA (97.3% vs. 83.8%) and GA (94.1% vs. 85.3%). These improvements are consistent with 3D-Belief providing an explicit and more geometrically grounded belief update, which better supports motion- and space-related inference under viewpoint changes. Meanwhile, performance on OM and Pers is not the best among methods (e.g., OM 73.9% vs. 78.3%, Pers 75.8% vs. 84.8%), suggesting that fine-grained object-centric dynamics remain challenging and may require stronger object-level temporal modeling or higher-fidelity appearance tracking.

A4. Implementation Details

A4.1. Model Training

A4.1.1. Datasets

We train our model on a composite dataset, which consists of SPOC Ehsani et al. [2024], RealEstate10K Zhou et al. [2018], DL3DV Ling et al. [2024], and Habitat-Matterport 3D Ramakrishnan et al. [2021]. At each step, we randomly select a subset with equal weights and sample frames. Each sample includes the context view and the target view with randomly spaced frames. To align the datasets, we set the minimum interval to 2. The maximum intervals for SPOC, RealEstate10K, DL3DV, and Habitat-Matterport 3D are 10, 100, 10, and 15, respectively.

Table A6: Extended results with VLMs for semantic representations in simulated object navigation tasks.

Models	**SR%** $\uparrow$	**SPL%** $\uparrow$	**SEL%** $\uparrow$	**Token (/step)** $\downarrow$
3D-Belief	45.83	36.43	32.99	0
w/o semantics (w/ GPT-5m)	49.58	40.10	41.01	79.42
w/o semantics (w/ Gemini 3)	46.22	36.34	38.02	340.67

Table A7: Results on Object Completion across different visibility.

Models	**Visibility**	**BEV IoU**$\uparrow$	**3D IoU**$\uparrow$	**Chamfer**$\downarrow$	**SigLIP**$\uparrow$	**Recognition**$\uparrow$
DFoT-VGGT	0.05	0.110	0.064	2.681	0.265	0.126
	0.55	0.362	0.243	0.830	0.798	0.767
	0.95	0.372	0.242	0.189	0.857	0.838
3D Belief	0.05	**0.147**	**0.083**	**2.435**	**0.329**	**0.165**
	0.55	**0.484**	**0.318**	**0.216**	**0.855**	**0.930**
	0.95	**0.535**	**0.369**	**0.187**	**0.884**	**0.909**

A4.1.2. Training Settings

3D-Belief The U-ViT backbone Hoogeboom et al. [2023, 2024] used to encode input images and the depth predictor used to estimate Gaussian primitives are initialized from pretrained weights: the DFoT encoder Song et al. [2025] and the MVSplat multi-view depth predictor Chen et al. [2024], respectively, both pretrained on RealEstate10K Zhou et al. [2018]. We use AdamW optimizer with a learning rate of 2×10^{-5} and weight decay of 0.001 for training, applying linear warm-up in the beginning 10K steps followed by cosine decay. The model can still be trained effectively and converge even with a small batch size. We set the batch size to 1 and trained for a total of 250K steps on a single H100 80GB chip. We follow DFM Tewari et al. [2023] to apply loss on the denoised target view image. We calculate L1 loss and LPIPS loss with a weight of 1.0 between the predicted image and the ground-truth. Although deep supervision is not employed, the model incorporates a depth smoothness loss with a weight of 0.1 as a regularization term. Additionally, the semantic loss based on CLIP Ilharco et al. [2021] and DINOv3 Siméoni et al. [2025] is calculated separately, with a weight of 0.1. Besides the target view, we also compute all the aforementioned loss for intermediate frames to enhance consistency. To further enhance the model's capabilities, we align the prediction and ground truth at the feature level with a weight of 5.0, while simultaneously aligning the prediction features using the pre-trained VGGT Wang et al. [2025a] model with a weight of 2.0. Training is completed by performing a weighted sum of all losses.

DFoT We use the DFoT model Song et al. [2025] trained on RealEstate10K Zhou et al. [2018] as the base model, and finetune it on the SPOC Ehsani et al. [2024] dataset with light horizontal flip augmentation. Training uses AdamW with learning rate 1×10^{-5}, weight decay 0.01, per-GPU batch size 4 with gradient accumulation of 2, and runs for 48 epochs on $4 \times$ H100 80GB chips. The diffusion process follows a cosine schedule with sigmoid loss weighting.

NWM We use the official pretrained NWM Bar et al. [2025] CDiT/XL model, which covers four robotics datasets (SCAND Karnan et al. [2022], TartanDrive Triest et al. [2022], RECON Shah et al. [2021], and HuRoN Hirose et al. [2023]) and Ego4D Grauman et al. [2022] videos. Then, we finetune it

Table A8: Results on Room Completion.

Models	Obj. Precision ↑	Obj. Recall ↑	Obj. F1 ↑	Occ. Acc. ↑	IoU Free ↑	IoU Occupied ↑	Occ. IoU ↑
DFoT-VGGT	0.639	**0.516**	0.531	0.252	0.104	0.115	0.110
3D Belief	**0.678**	0.490	**0.536**	**0.900**	**0.648**	**0.235**	**0.442**

Table A9: Results on SAT-Real.

	SAT Real					
	Avg	EM	OM	EA	GA	Pers
Gemini-3.0	85.3	95.7	**78.3**	83.8	85.3	**84.8**
+ SWM	86.7	95.0	70.0	94.1	89.7	81.3
+ SVC	86.0	95.7	**78.3**	86.5	**94.1**	75.8
+ **3D Belief**	**88.7**	**100.0**	73.9	**97.3**	**94.1**	75.8

on SPOC Ehsani et al. [2024] dataset using AdamW optimizer with a learning rate of 8×10^{-5} and a batch size of 16, and a total of 200K training steps. Training is done on a single H100 80GB chip.

VGGT We utilize the official pretrained VGGT-1B model Wang et al. [2025a] with both camera and depth heads enabled. We finetune it on the SPOC Ehsani et al. [2024] dataset on 238×238 inputs (patch size 14) using AdamW (learning rate 1×10^-5, weight decay 0.05) for 60 epochs with an effective batch size of 96. Training is done on $4 \times$ GH200 96GB chips.

SPOC We finetune the pretrained SigLIP-ViTb-3 CHORES-Nav model Ehsani et al. [2024] (an EarlyFusionCnnTransformer policy) on the SPOC Ehsani et al. [2024] RERENDERED mixture, which combines all four ObjectNav task types (ObjectNavType, ObjectNavRoom, ObjectNavRelAttribute, and ObjectNavLocalRef). The policy consumes raw navigation and manipulation camera streams together with the agent's last-action and in-hand-object signals. Training is performed on $4 \times$ H100 80GB chips with a per-GPU batch size of 6 and a sliding window of 50 steps, using a learning rate of 2×10^{-5} and mixed-precision (fp16) for 50 epochs over a 100,000-sample subset of the dataset.

A4.2. Planning

A4.2.1. Object Navigation Tasks

We provide additional implementation details for the object navigation experiments. At each step, the agent updates its 3D belief z^t from the latest egocentric RGB stream and poses. A frontier goal sampler then proposes the next possible waypoint goals, computes candidate paths with an A* planner, and performs mental simulation by rendering imagined observations along each path from z^t. These rollouts are scored by goal progress and information gain via semantic-map queries. The agent executes a short prefix of the best plan and repeats.

We describe each key component below.

Waypoint Sampling. Planning directly to the final goal is difficult under partial observability, since large portions of the environment are unseen and therefore uncertain. We instead plan an intermediate waypoint at each step. Waypoints are selected in two cases. If the final goal (a target object or a target location) has already appeared in the observed history $o^{1:t}$, we set the waypoint to the estimated goal position by querying the semantic map. Otherwise, we sample a small set of candidate waypoints in currently unobserved regions within an exploration radius centered at the agent's current position.

Simulating and Evaluating the Paths. For each of the waypoints, we can sample a path and simulate the agent's imagined observations along that path using the predicted scene representation z^t. These rollouts estimate how informative or goal-directed a path is. For example, whether the target object becomes visible in the imagined views or whether the rendered semantics indicate proximity to the goal. Specifically, we score each path by querying the rendered semantic feature maps to assess progress and promise from the imagined observations. If the semantics map indicates that it is likely that the object is on the imagined path, then we will score the path highly.

Execution. We execute the first T steps of the highest-scoring plan, then re-estimate the scene representation from the newly acquired observations and repeat the procedure iteratively until the goal is reached.

In all planning experiments, we use 3 hypotheses for 3D-Belief sampling at each decision step. The results for the 3D-Belief model and SPOC are obtained using a single H100 80GB chip for inference, while other evaluations rely on a single A100 80GB chip.

A4.2.2. Real-World Experiments

For the object navigation task, we select 5 target objects and randomize each episode by (1) sampling a target object and (2) initializing the robot from one of 3 predefined starting locations to induce different exploration requirements. An example setup is shown in Fig. A3.

We use the Stretch 3 Mobile Manipulator (Hello Robot) as our real-world embodied platform. The 3D-Belief model and all baselines run on an external GPU client (an NVIDIA RTX 4090 desktop), and the robot communicates with the client via ROS 2.

We mount an RGB-D camera (*Intel RealSense* D455) on the robot and use the RGB stream as the visual input. The robot's pose is estimated using onboard wheel-encoder odometry. Together, the RGB observations and associated poses form an egocentric stream that serves as input to our models. Note that 3D-Belief uses only RGB at inference time. However, the real-world vision datasets used for training provide camera poses from *Structure-from-Motion (SfM)*, which are defined only up to an unknown global scale; as a result, the model's depth predictions are not inherently metric. To recover metric-consistent geometry for planning, we align the predicted depth map $\hat{d}$ to the sensed depth d by estimating a per-sequence scalar scale s via robust regression, and apply this scale to the depth maps produced by the Multi-view Depth Predictor (Sec. 3.2).

A4.3. Reasoning

A4.3.1. 3D-CORE Metrics

For *Object Completion*, BEV IoU, 3D IoU, and Chamfer distance are calculated by the extracted point cloud of the imagined object (segmented by Gemini 2.5 Comanici et al. [2025]) and the GT point cloud

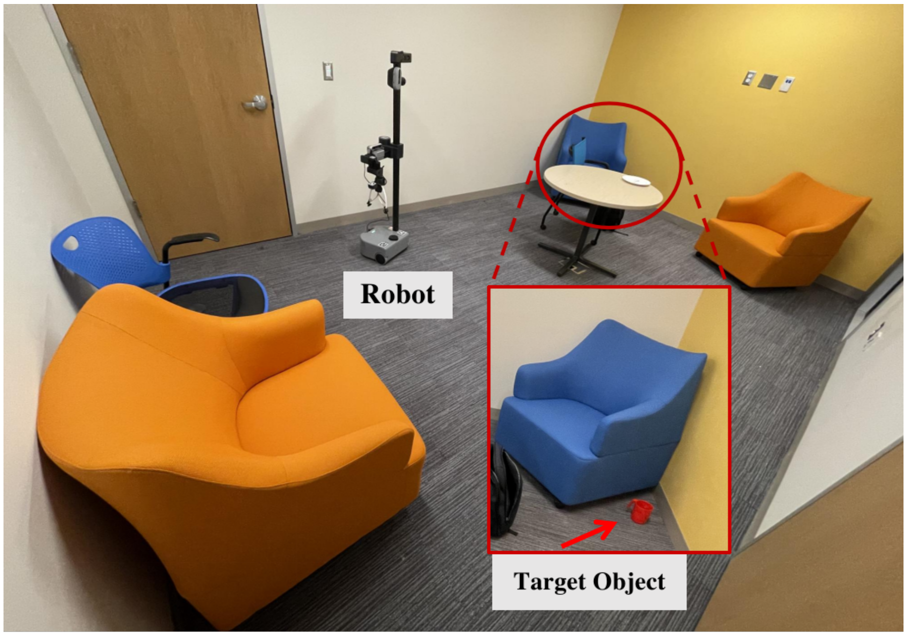

Figure A3: Example setup for the real-world object navigation tasks with 3D-Belief. The small image shows a zoom-in view near the blue couch with a target object hidden.

(segmented by GT masks). SigLIP is calculated by comparing the predicted and GT views at the same camera poses. Recognition is defined by the success rate of target object recognition by VLMs in the imagined views.

For *Room Completion*, Obj. F1 is calculated by comparing a list of VLM (Gemini 2.5 Comanici et al. [2025]) recognized objects in imagined views, with the GT object list in that view. Occ. Acc. is defined by the accuracy on cells where both gt and pred are known (0 or 1). IoU Free is the IoU for free cells (class 0), IoU Occupied is the IoU for occupied cells (class 1), Occ. IoU is the average of the previous two. For *Object Permanence*, LPIPS and SigLIP are calculated using the first and the last rendered image, whose camera poses are the same. All reasoning experiments utilize a single A100 80GB chip for inference.

A5. Discussion

How does 3D imagination affect 2D rendering quality?

While 3D-Belief is designed for explicit 3D reasoning, it also improves *2D* rendering quality as shown in Section 4.1. Because predictions are structured in an explicit 3D scene, generated frames are constrained by coherent geometry instead of being synthesized purely in pixel space. This acts as a strong prior for cross-view consistency: once surfaces and free space are established in 3D, new viewpoints are rendered by projection, reducing viewpoint ambiguity and limiting texture drift or identity switches over long horizons. This yields better perceptual/distributional metrics (lower LPIPS/FID/FVD) and higher PSNR/SSIM. Qualitatively (Appendix A3.1.1), 2D-only baselines often drift under large camera motion, whereas 3D-Belief better preserves rigid structure and object permanence across revisits.

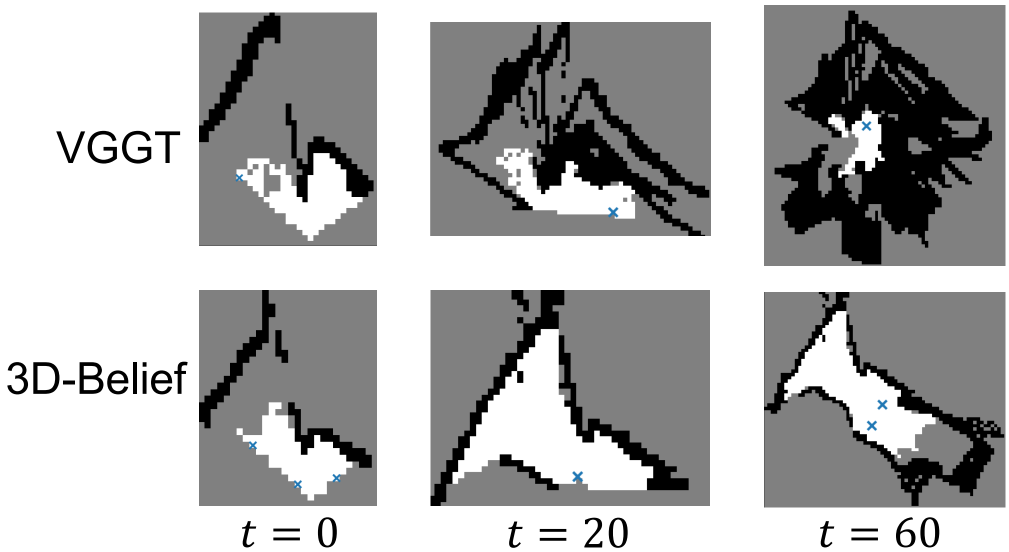

Figure A4: Comparison of inferred occupancy map generated by 3D-Belief and VGGT (w/ frontier) over time in the same task.

How do the key 3D belief model capacities affect the embodied task performance?

A key advantage of 3D-Belief is maintaining a spatially consistent scene memory over long rollouts. In Figure A4, VGGT-based occupancy quickly accumulates duplicated structure, and maps often break afterwards, trapping the agent. In contrast, 3D-Belief remains coherent and steadily expands over much longer horizons, supporting continued exploration and replanning. This failure mode is worse for DFoT-VGGT and NWM-VGGT, where both lifting and video prediction drift under long horizons, corrupting the 3D cache and degrading planning Xiao et al. [2025], Melnik et al. [2024], Oshima et al. [2025].

Table A4 in Appendix A3.2.2 further shows that removing multi-hypothesis sampling causes a performance drop, indicating that maintaining multiple plausible beliefs is important for robust decision making. Together with internal semantic prediction, this lets the agent compare candidate actions conditioned on task or language specifications, rather than relying mainly on waypoint sampling. It also reduces deployment cost: compared to attaching an external VLM for semantics, which adds token and compute overhead (Table 3), 3D-Belief provides semantics within the model.

A6. Broader Impacts

3D-Belief aims to improve the ability of embodied agents to reason about partially observed environments by maintaining and updating explicit 3D beliefs over both observed and unobserved space. It has potential benefits for assistive robotics, object search, warehouse inspection, and navigation in areas like homes or hospitals. By representing uncertainty explicitly and supporting multi-hypothesis reasoning, it may help agents explore more efficiently and make better-informed decisions.

However, predicting unobserved regions also introduces risks. Plausible but incorrect beliefs could lead to unsafe robot behavior if treated as facts. In indoor settings, explicit 3D scene representations may also capture sensitive information about private spaces and personal belonging, raising privacy and surveillance concerns. This work should therefore be viewed as a research step rather than a deployment-ready system. Practical deployments should include conservative planning, collision checking, privacy-preserving data handling, and clear separation between observed evidence and imagined scene completions.

A7. Asset Licenses

This section documents the licenses and terms of use for existing assets used in the paper. We do not claim ownership of these third-party assets, and we comply with their corresponding license and usage requirements.

Table A10: Existing assets used in this work and their licenses or terms of use.

Asset	**License / Terms**
AI2-THOR Kolve et al. [2017]	Apache-2.0
ProcTHOR / ProcTHOR-10K Deitke et al. [2022]	Apache-2.0
Objaverse Deitke et al. [2023]	ODC-By v1.0 dataset license; per-object Creative Commons licenses
RealEstate10K Zhou et al. [2018]	CC-BY 4.0
OpenCLIP Ilharco et al. [2021]	MIT
DINOv3 Siméoni et al. [2025]	DINOv3 License
DFoT Song et al. [2025]	MIT
Navigation World Models (NWM) Bar et al. [2025]	CC-BY-NC 4.0
VGGT Wang et al. [2025a]	Meta Research Materials License
Qwen3-VL-8B-Instruct Yang et al. [2025a]	Apache-2.0